Ref. [52] — Dropping Convexity for Faster Semi-definite Optimization

Srinadh Bhojanapalli, Anastasios Kyrillidis, Sujay Sanghavi | COLT 2016

1. The question: why optimize a non-convex factorization?

Many problems ask us to minimize a *convex* function $f(X)$ under the constraint $X \succeq 0$. A positive-semidefinite $n \times n$ matrix can be expressed as $X = UU^\top$ for $U \in \mathbb{R}^{n\times r}$. This reduces storage when $r \ll n$, but the map from U to $f(UU^\top)$ becomes non-convex. Can gradient descent on the factors still converge rapidly?

2. The method: factored gradient descent

Write the new objective $g(U) = f(UU^\top)$. Assuming differentiability and a symmetric gradient of f with respect to X, differential calculus gives

$$dX = dU\, U^\top + U\, dU^\top, \quad dg = \langle \nabla f(X), dX \rangle, \quad \boxed{\nabla g(U) = 2\nabla f(UU^\top)U.}$$

Therefore the factor-based gradient step becomes

$$\boxed{U_{k+1} = U_k - 2\eta_k \nabla f(U_k U_k^\top)U_k.}$$

For the teaching example $f(X) = \frac{1}{2}\|X - X_*\|_F^2$, the factor gradient is

$$\nabla g(U) = 2(UU^\top - X_*)U.$$

Although f is convex, g is generally not. For instance, U and UQ define the same PSD matrix for every orthogonal Q, so the factor problem has symmetries and non-isolated solutions.

3. Precisely what the authors prove

The authors design a stepsize rule for *factored gradient descent* (FGD) based on assumptions on the original convex objective f. They prove *local* convergence rates mirroring the ordinary convex rates:

$$\text{smooth } f: \quad O(1/k), \qquad \text{(restricted) strongly convex } f: \quad \text{exponential/linear convergence in } k.$$

For the restricted strongly convex setting, they also give an initialization procedure using first-order oracle access to f; for several problem instances, pairing this with FGD provides global convergence assurances. The distinctions between *local convergence* and *global convergence after suitable initialization* are essential.

Paper result. Factorization can trade the original convex geometry for a lower-dimensional non-convex problem without losing local convergence rates under suitable conditions. The analysis provides an algorithmic route, not a theorem that every critical point of every factorized objective is globally optimal.

4. How this links to neural networks

A two-layer linear network computes W_2W_1x; PSD factorization uses the multiplicative parameterization $UU^\top$. Both create non-convexity by writing a matrix as a product. The comparison helps explain why non-convex *parameterizations* need not make optimization hopeless. However, the constraints and model classes differ: PSD optimization is not generic deep learning.

5. Assumptions, limits, survey relevance

The theorems assume smoothness, appropriate convexity or restricted strong convexity of f, stepsize control, and—in the global guarantees—appropriate initialization. Rank selection matters. The paper does not give a universal strict-saddle classification for neural-network losses, nor an efficient SOSP result for arbitrary g. Use primarily for **RQ1** (benign factorized geometry) and **RQ3** (computationally economical optimization), with the structured-versus-realistic-model caution of **RQ4**.

Do not overstate. "FGD converges locally at convex-like rates" is different from "all local minima are global"; the latter is a separate landscape question and is not the universal theorem of this work.

6. Our notes

This is a useful counterexample to the idea that non-convexity automatically makes gradient descent slow. By writing a PSD matrix as $UU^\top$, we give up convexity but save memory and can keep strong convergence rates with the right initialization and assumptions. It is a structured matrix problem, so we should use it as a bridge to neural-network factorization rather than a claim about every neural network. **Original source:** https://proceedings.mlr.press/v49/bhojanapalli16.html. **Verification:** official COLT abstract and methodology checked; displayed gradients are transparent teaching derivations.